

DSC 465- Homework 4

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- 1) Create a visualization using one of the techniques from the latter half of the class (after the midterm). For example, from Week 7 you could use one of the techniques from the Categorical unit, like a mosaic plot or Bertin matrix, or you could try applying an interactivity example to your project data. Week 8 offers uncertainty visualizations and contours and 2D binning to apply to numerical variable relationships. Despite being covered earlier, a cartogram is also allowed. Each group member's visualizations must be distinct. The group's visualizations can use the same technique if they cover different aspects of the data or use the technique in different ways.

```
> rm(list = ls())
> library(ggplot2)
> library(ggmosaic)
> police_killings<- read.csv(file.choose(), header = TRUE)
> head(police_killings)
```

	name	age	gender	raceethnicity	month	day	year
1	A'donte Washington	16	Male	Black	February	23	2015
2	Aaron Rutledge	27	Male	white	April	2	2015
3	Aaron Siler	26	Male	white	March	14	2015
4	Aaron Valdez	25	Male	Hispanic/Latino	March	11	2015
5	Adam Jovicic	29	Male	white	March	19	2015
6	Adam Reinhart	29	Male	white	March	7	2015

	streetaddress	city	state	latitude	longitude	state_fp	county_fp
1	Clearview Ln	Millbrook	AL	32.52958	-86.36283	1	51
2	300 block Iris Park Dr	Pineville	LA	31.32174	-92.43486	22	79
3	22nd Ave and 56th St	Kenosha	WI	42.58356	-87.83571	55	59
4	3000 Seminole Ave	South Gate	CA	33.93930	-118.21946	6	37
5	364 Hiwood Ave	Munroe Falls	OH	41.14857	-81.42988	39	153
6	18th St and Palm Ln	Phoenix	AZ	33.46938	-112.04332	4	13

	tract_ce	geo_id	county_id	name1sad	lawenforcementagency
1	30902	1051030902	1051	Census Tract 309.02	Millbrook Police Department
2	11700	22079011700	22079	Census Tract 117	Rapides Parish Sheriff's Office
3	1200	55059001200	55059	Census Tract 12	Kenosha Police Department
4	535607	6037535607	6037	Census Tract 5356.07	South Gate Police Department
5	530800	39153530800	39153	Census Tract 5308	Kent Police Department
6	111602	4013111602	4013	Census Tract 1116.02	Phoenix Police Department

	cause	armed	pop	share_white	share_black	share_hispanic	p_income	h_income
1	Gunshot	No	3779	60.5	30.5	5.6	28375	51367
2	Gunshot	No	2769	53.8	36.2	0.5	14678	27972
3	Gunshot	No	4079	73.8	7.7	16.8	25286	45365
4	Gunshot Firearm	4343		1.2	0.6	98.8	17194	48295
5	Gunshot	No	6809	92.5	1.4	1.7	33954	68785
6	Gunshot	No	4682	7.0	7.7	79.0	15523	20833

	county_income	comp_income	county_bucket	nat_bucket	pov	urate	college
1	54766	0.9379359		3	14.1	0.09768638	0.16850951
2	40930	0.6834107		2	1	28.8	0.06572379
3	54930	0.8258693		2	3	14.6	0.16629314
4	55909	0.8638144		3	3	11.7	0.12482727
5	49669	1.3848678		5	4	1.9	0.06354983
6	53596	0.3887044		1	1	58.0	0.07365145

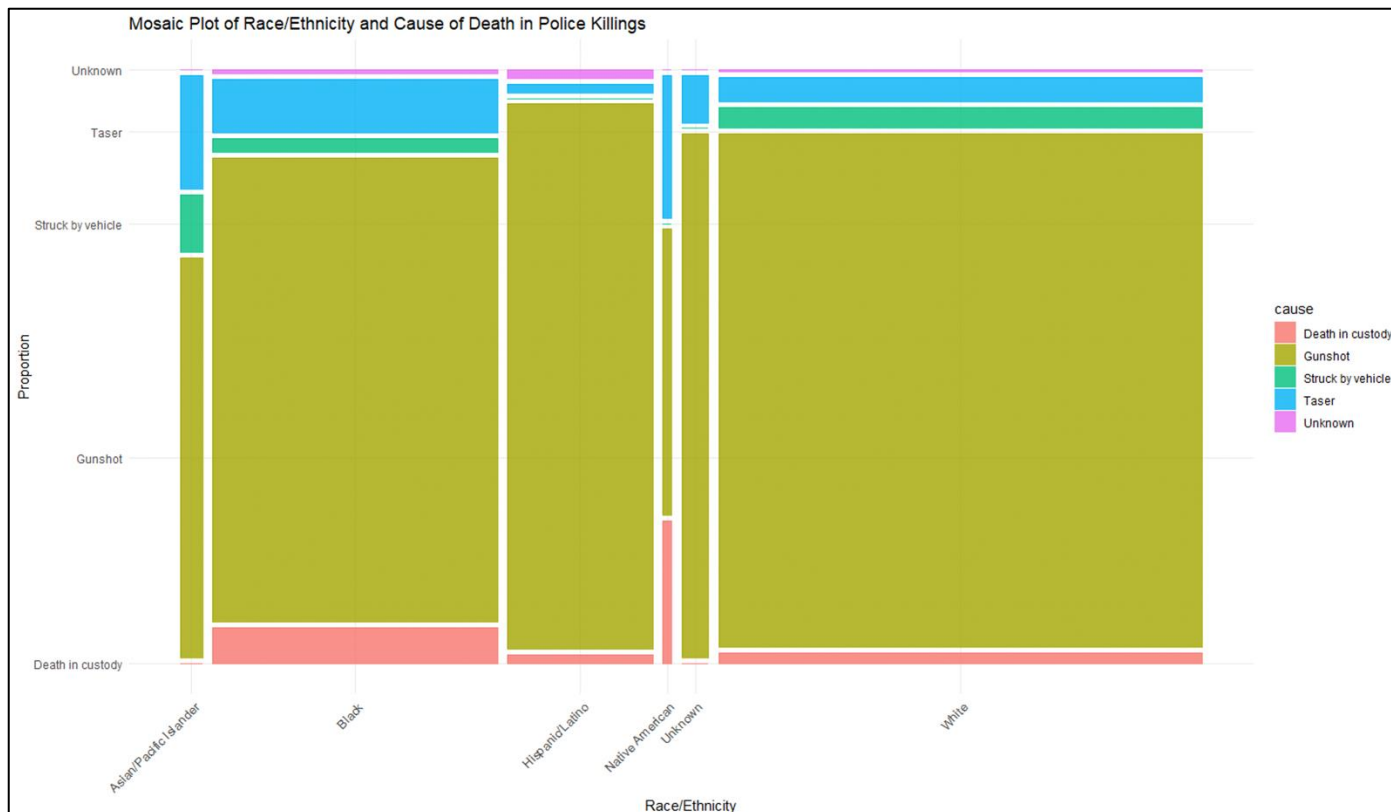
	date
1	2015-02-23
2	2015-04-02

```

      date
1 2015-02-23
2 2015-04-02
3 2015-03-14
4 2015-03-11
5 2015-03-19
6 2015-03-07
> str(police_killings)
'data.frame': 436 obs. of 35 variables:
 $ name      : chr "A'donte washington" "Aaron Rutledge" "Aaron Siler" "Aaron valdez" ...
 $ age       : int 16 27 26 25 29 29 22 35 44 31 ...
 $ gender    : chr "Male" "Male" "Male" "Male" ...
 $ raceethnicity : chr "Black" "white" "white" "Hispanic/Latino" ...
 $ month     : chr "February" "April" "March" "March" ...
 $ day       : int 23 2 14 11 19 7 27 26 28 7 ...
 $ year      : int 2015 2015 2015 2015 2015 2015 2015 2015 2015 ...
 $ streetaddress : chr "Clearview Ln" "300 block Iris Park Dr" "22nd Ave and 56th St" "3000 Seminole Ave" ...
 $ city      : chr "Millbrook" "Pineville" "Kenosha" "South Gate" ...
 $ state     : chr "AL" "LA" "WI" "CA" ...
 $ latitude  : num 32.5 31.3 42.6 33.9 41.1 ...
 $ longitude : num -86.4 -92.4 -87.8 -118.2 -81.4 ...
 $ state_fp  : int 1 22 55 6 39 4 6 6 48 26 ...
 $ county_fp : int 51 79 59 37 153 13 29 37 41 81 ...
 $ tract_ce  : int 30902 11700 1200 535607 530800 111602 700 294200 603 14200 ...
 $ geo_id    : num 1.05e+09 2.21e+10 5.51e+10 6.04e+09 3.92e+10 ...
 $ county_id : int 1051 22079 55059 6037 39153 4013 6029 6037 48041 26081 ...
 $ namesad   : chr "Census Tract 309.02" "Census Tract 117" "Census Tract 12" "Census Tract 5356.07" ...
 $ lawenforcementagency: chr "Millbrook Police Department" "Rapides Parish Sheriff's Office" "Kenosha Police Department"
 "South Gate Police Department" ...
 $ cause     : chr "Gunshot" "Gunshot" "Gunshot" "Gunshot" ...
 $ armed     : chr "No" "No" "No" "Firearm" ...
 $ pop       : int 3779 2769 4079 4343 6809 4682 5027 5238 4832 3795 ...
 $ share_white : num 60.5 53.8 73.8 1.2 92.5 7 50.8 8.6 14.6 63.6 ...
 $ share_black : num 30.5 36.2 7.7 0.6 1.4 7.7 0.3 0.2 17.7 7.7 ...
 $ share_hispanic : num 5.6 0.5 16.8 98.8 1.7 79 44.2 84.1 66.3 26.5 ...
 $ p_income   : int 28375 14678 25286 17194 33954 15523 25949 25043 16778 22005 ...
 $ h_income   : int 51367 27972 45365 48295 68785 20833 58068 66543 30391 44553 ...
 $ county_income : int 54766 40930 54930 55909 49669 53596 48552 55909 38310 51667 ...
 $ comp_income : num 0.938 0.683 0.826 0.864 1.385 ...
 $ county_bucket : int 3 2 2 3 5 1 4 4 2 3 ...
 $ nat_bucket  : int 3 1 3 3 4 1 4 4 1 2 ...
 $ pov        : num 14.1 28.8 14.6 11.7 1.9 58 17.2 12.2 37.7 18.4 ...
 $ urate      : num 0.0977 0.0657 0.1663 0.1248 0.0635 ...

 $ college    : num 0.1685 0.1114 0.1473 0.0501 0.404 ...
 $ date       : chr "2015-02-23" "2015-04-02" "2015-03-14" "2015-03-11" ...
> # Generate a mosaic plot using 'raceethnicity' and 'cause'
> ggplot(data = police_killings) +
+   geom_mosaic(aes(x = product(raceethnicity), fill = cause)) +
+   labs(title = "Mosaic Plot of Race/Ethnicity and Cause of Death in Police Killings",
+        x = "Race/Ethnicity",
+        y = "Proportion") +
+   theme_minimal() +
+   theme(axis.text.x = element_text(angle = 45, hjust = 1, size = 10)) # Adjust x-axis label formatting

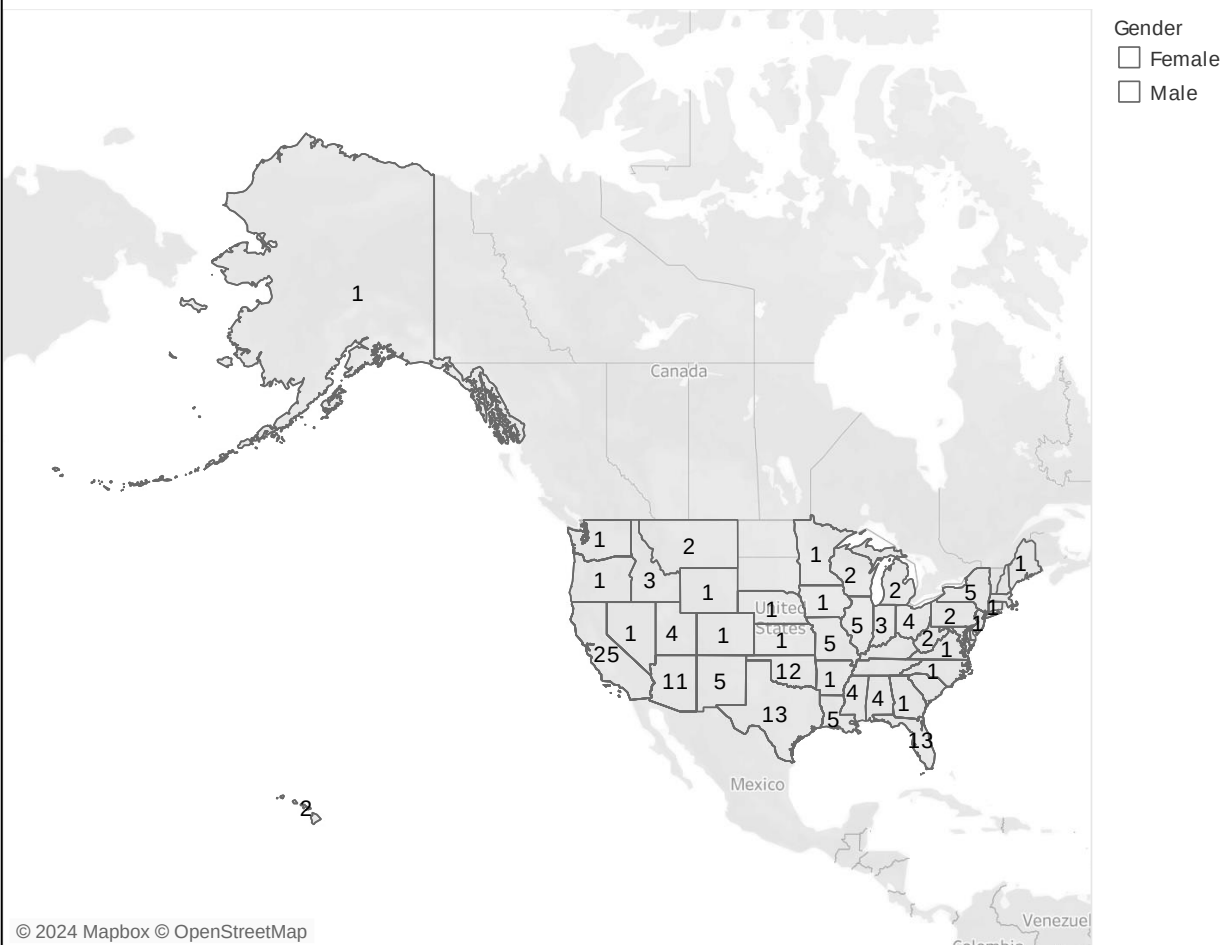
```



The mosaic plot visualizes the relationship between **race/ethnicity** and the **cause of death** in police killings. It shows that **gunshot** (in green) is the most common cause of death across all racial groups, particularly in the **Black** and **White** populations. Other causes, like **death in custody** (red) and **taser** (blue), are less frequent. The **Unknown** category (purple) appears sporadically but is more prevalent in certain racial groups.

- 2) Do the same as in item 1 but for another type of visualization. For this, you may use visualizations going back to Week 4, which includes geographical, statistical and special timeseries plots (e.g. tile plots or line graphs with smoothing). You may use the same type of visualization as item 1 if the two cover different aspects of the data or use the technique in different ways.

Geographical Distribution of Police Killings by Gender & Race/ Ethnicity in the U.S



Map based on Longitude (generated) and Latitude (generated). Color shows details about Gender. The marks are labeled by count of cleaned_police_killings.csv. Details are shown for State and Race/ethnicity.

This geographical map represents the distribution of police killings in the U.S. broken down by gender and race/ethnicity. The color coding shows gender (orange for males and blue for females), while the tooltips provide details about race/ethnicity and the number of police killings in each state.

Males Predominate: In almost every state, males (orange) make up the majority of police killings, as indicated by the overwhelming presence of orange shading on the map.

Female Involvement: Some states show instances of police killings involving females (blue), but these are significantly fewer in number, as seen in states like California, Georgia, and Texas.

Race/Ethnicity Information: The tooltip provides detailed information on the race/ethnicity of the individuals involved. For example, in Alaska, the one recorded police killing was of a Hispanic/Latino male.

High-Impact States: States like California (25 incidents) and Georgia (13 incidents) have higher counts of police killings. This indicates regions with more frequent incidents.